appropriate base pairs. The process results in two strands of whole DNA that are exactly the same as the original.

Since DNA remains in the nucleus of the cell, a related molecule called messenger ribonucleic acid (mRNA) copies segments of DNA coding sequence and carries the information to the regions of the cell where new proteins are made. RNA is chemically related to DNA, but the thymine base (T) is replaced by the base uracil (U), which is less stable but requires less energy to make. Stable living organisms benefit from having DNA genomes, but RNA makes up genomes of some viruses, where stability can be less advantageous.

DNA is found in all living things on Earth, from amoebae to insects, to trees, tigers, and humans. Of course, the sequence of base pairs varies, and this difference allows geneticists to trace relationships between different species.